1

$2y' = \frac{1}{y}$ $\quad y=0 \to$ stac. řešení

$\int y\,dy = \int \frac{1}{2}\,dx$

$y^2 = x + 2C$

$\cancel{y = \sqrt{x+D}}$

$y = \pm\sqrt{x+D}$

stac. řešení zahrnuto v případech $D = -x$

$\cancel{y = \sqrt{x+D}}$; $\cancel{x+D \geq 0}$

$x + D \geq 0$

$x \geq (-D)$

$$\boxed{y = \pm\sqrt{x+D}\ ;\ \text{pro } x, y \in \mathbb{R} \text{ a musí platit } x \geq (-D)}$$

a)

$y(1) = 2$

$2^2 = 1 + D$

$4 = 1 + D$

$D = 3$

b)

$y(2) = -1$

$(-1)^2 = 2 + D$

$1 = 2 + D$

$D = (-1)$

c) $y(-1) = 3$

$3^2 = -1 + D$

$10 = D$

2)

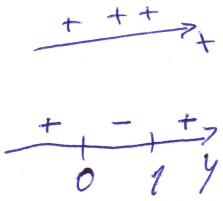
$\cancel{y=0 \to stac}$

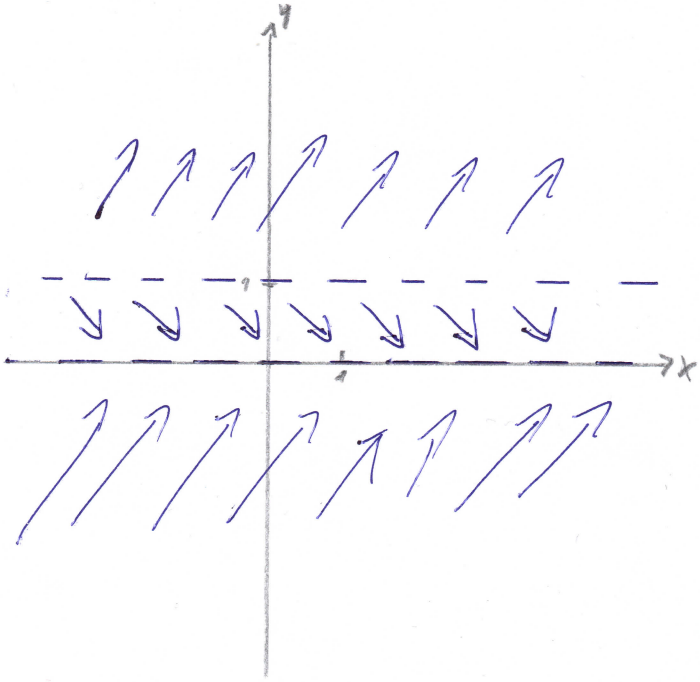
$y' = y^4 - y^3$

$0 = y^4 - y^3$

$y = 0 \to$ stac. řešení

$y = 1 \to$ stac. řešení





$y = 1 \to$ nestabilní

$y = 0 \to$ stabilní

③

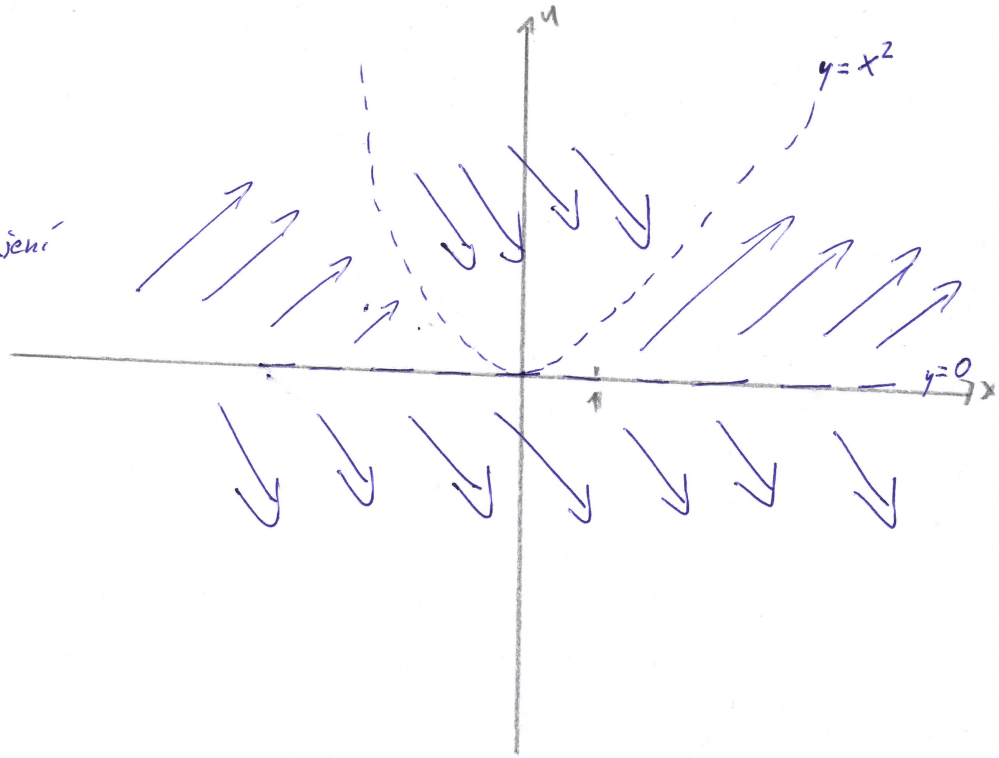
$y' = y(x^2 - y)$

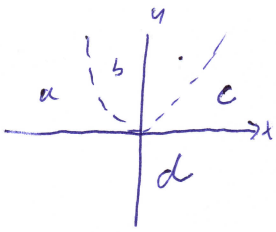
$0 = y(x^2 - y)$

$y = 0 \quad (x^2 - y) = 0$

$x^2 = y$

$y = 0 \rightarrow$ stac. řešení





a) $(-2, 2)$

$y' = 2 \cdot (4 - 2)$

$y' = 2 \cdot 2$

$y' = 4$ (+) ne

b) $(0, 1)$

$y' = 1(0 - 1)$

$y' = 1 \cdot (-1)$

$y' = (-1)$ (−)

$a) = d)$

d) $(0, -1)$

$y' = (-1)(0 + 1)$

$y' = (-1)$ (−)